Supplementary Material for "A Dynamic-Scene Gaussian SLAM Map: Anatomy, Compression, and Offline Repair"

Anonymous submission. This supplement provides the per-system, per-sequence and per-map detail behind the main paper's summary tables and prose. Every number is generated by scripts from the same logged artifacts as the main paper; no new experiments were run. Tables S1–S8 and Fig. S1 are referenced in the main text where noted.

S1 Reproduction detail (main paper §3)

Protocol. All eleven reproduced systems were run on the same machine under one protocol: identical evaluation harness, Horn SE(3) alignment, ATE RMSE in centimetres, three seeds, all frames. Diverged runs (path length $> 5\times$ GT or ATE > 100 cm) are flagged and re-run under the same protocol with recommended settings; the flag is kept in the table. DynaGSLAM delegates tracking to an external front-end (DynoSAM poses), so its seeds are deterministic and it is reported without variance. One reproduced system, WildGS-SLAM, is monocular and receives RGB only; every other system receives RGB-D.

Coverage. Tracking is covered on all eight BONN sequences for every method. Rendering metrics exist only for methods with a rendering path: DynaSLAM, NGD-SLAM and ORB-SLAM3 have none, so they are omitted from the rendering tables. Table S1 reports the full reproduction on the eight BONN sequences, including the two sequences (*mv_no_box*, *mv_no_box2*) that the main paper's Table 1 excludes because the prior stack has no run on them; the six-sequence columns are the comparable subset. Table S2 reports the prior stack on its seven BONN sequences, including *crowd3*, which no baseline ran. The stack is evaluated on the same machine and protocol in an independent batch and is reported as background.

S2 Codec detail (main paper §5)

Per-sequence working points. The three fixed bit-width working points on the 42 frozen maps (seven sequences $\times$ two arms $\times$ three seeds), mean per sequence. The held-out panel (36 unseen maps, five TUM sequences and one Bonn, *moving_obstructing_box*) reproduces the near-lossless point with per-map ratios spanning 5.72–6.12 (mean 5.96) at -0.01 dB, agreeing with the frozen-map point of the main paper's Table 4.

Learned baselines, per-configuration detail. SOG (Morgenstern et al., ECCV 2024) runs its official non-sorted path on the frozen PLYs, both its unquantised configuration and its quantised configuration; the gap between them is the evidence that SOG's compression gain is dominated by quantisation, matching the main paper's finding for the parameter-free codec. The Niedermayr c3dgs row is a refit (QAT fine-tuning from the frozen map); its reference implementation fixes no random seed, so per-map values are not reproducible beyond $\approx$0.1 dB, and the row is reported without a variance claim.

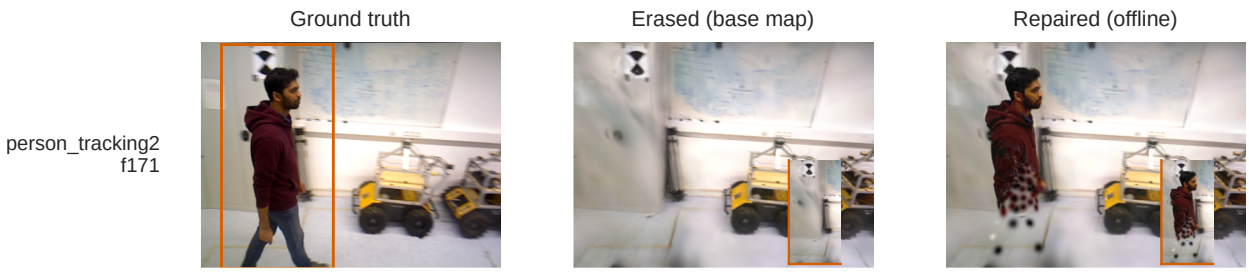


Figure S1: Offline repair on the third repair sequence (*person_tracking2*), one deterministic quantile frame, rendered at the evaluated frame's own pose. Columns: ground truth; base map with dynamic content erased; same base map with the dynamic layer restored. As in the main paper's Fig. 6, the erased content returns without touching the static map; individual inserted primitives are visible at close zoom, a boundary of this repair path.

S3 Anti-dynamic cost per map (main paper §6)

Per-map values. Each row is one frozen map (sequence $\times$ arm); the cost of erasing dynamic content, the recoverable fraction under a static insertion, and the static-side effect. These are the per-map values behind the interval reported in the main text; the interval is given instead of a mean because the distribution is sign-mixed on the static side.

S4 Repair, per sequence and seed (main paper §7)

Per-cell values. The locked criteria are J1 ≥ 2.5 dB, J2 ≤ 0.70, J3 within ± 0.25 dB; every sequence-and-seed cell passes all three. Values are samepose medians as in the main paper.

S5 Additional qualitative renders (main paper §7)

Method	balloon	balloon2	crowd	crowd2	person_tracking	person_tracking2	mv_no_box	mv_no_box2
Static methods								
MonoGS	39.32±1.01	22.05±1.55	86.47±18.34[d]	147.46±39.03[d]	44.83±9.06	43.85±8.55	15.33±9.47	16.84±16.96
SplaTAM	36.50±0.53	37.47±2.87	135.37±36.15[d]	493.22±470.66[d]	92.52±18.93[d]	132.72±9.37[d]	37.39±0.44	19.13±2.87
Co-SLAM	27.10±0.97	35.10±4.07	35866.32±38265.29[d]	24905.40±42817.31[d]	75.50±6.52	71.26±2.90	10.14±0.27	24.08±1.05
ORB-SLAM3	7.27±1.45	19.81±2.18	76.60±27.86[d]	111.41±14.75[d]	69.54±2.19	76.82±2.14	24.85±5.90	4.13±0.86
Dynamic methods								
DG-SLAM	3.66±0.02	3.83±0.08	5.76±0.15	6.39±0.50	4.25±0.06	6.12±0.13	3.01±0.09	3.56±0.02
RGD-SLAM	2.45±0.22	4.26±0.89	2.61±0.28	2.36±0.06	7.21±1.24	20.10±2.55	2.28±0.26	4.70±0.24
WildGS-SLAM	2.75±0.07	2.42±0.04	1.55±0.06	2.17±0.05	3.63±0.08	3.09±0.03	1.60±0.03	2.50±0.06
DynaGSLAM	8.36	9.36	23.82	13.73	8.94	13.69	12.11	17.17
DynaSLAM	2.88±0.16	2.59±0.10	2.07±0.21	2.85±0.28[d]	4.53±1.81	8.82±1.18	2.27±0.06	2.75±0.11
NGD-SLAM	2.82±0.07	2.46±0.08	1.85±0.16	2.22±0.07	4.97±0.44	6.63±0.43	2.03±0.08	3.07±0.23
RoDyn-SLAM	8.47±0.31	11.90±0.21	5.60±0.18	4.63±0.06	16.39±0.54	14.36±0.11	16.54±7.12	30.52±11.82

Table S1: Absolute trajectory error on all eight BONN sequences of the reproduction (RMSE, cm, mean±sd over 3 seeds). The two *mv_no_box* columns are baselines-only: the prior stack has no run on them, so they are outside the comparable subset of Table 1 in the main paper and are shown here for completeness. [d]: tracking diverged (path length $> 5\times$ GT or ATE > 100 cm).

Prior RGB-D stack	balloon	balloon2	crowd	crowd2	crowd3	person_tracking	person_tracking2
Prior RGB-D stack	3.25±0.07	5.48±0.27	2.23±0.10	2.26±0.32	1.53±0.02	11.31±0.70	10.40±0.28

Table S2: Absolute trajectory error of the prior RGB-D stack on its seven BONN sequences (RMSE, cm, mean±sd over 3 seeds; all-frames, Horn SE(3) alignment). The *crowd3* column is stack-only: no reproduced baseline has a run on it. Evaluated on the same machine and protocol in an independent batch; reported as background, not as a result of this paper.

Method	balloon	balloon2	crowd	crowd2	person_tracking	person_tracking2
MonoGS	19.30±0.24	19.17±0.12	15.84±0.77	16.51±1.43	17.81±0.65	19.68±0.69
SplaTAM	17.85±0.39	16.82±0.37	15.76±1.30	15.39±0.82	18.68±2.40	20.69±0.94
Co-SLAM	17.83±0.15	16.93±0.21	14.16±0.67	14.47±2.10	16.04±0.28	16.00±0.12
DG-SLAM	23.67±0.36	22.70±0.38	22.99±0.16	22.57±0.25	21.15±0.27	19.75±0.08
RGD-SLAM	25.14±0.18	24.36±0.18	24.21±0.26	24.36±0.06	24.11±0.06	22.76±0.34
WildGS-SLAM	18.31±0.09	16.52±0.07	16.43±0.09	15.64±0.13	18.74±0.19	18.58±0.12
DynaGSLAM	22.14±0.02	18.81±0.09	15.30±0.07	20.35±0.18	9.80±0.05	6.97±0.02
RoDyn-SLAM	15.89±0.04	14.22±0.25	15.56±0.01	14.31±0.12	15.67±0.40	14.99±0.02

Table S3: Full-image PSNR (dB, mean±sd over 3 seeds) of the reproduced systems on the common BONN sequences. DynaSLAM, NGD-SLAM and ORB-SLAM3 have no rendering path and are omitted (not fabricated). Same-machine, same protocol.

Method	balloon	balloon2	crowd	crowd2	person_tracking	person_tracking2
MonoGS	1.90±0.02	1.86±0.02	1.81±0.03	1.70±0.04	1.71±0.02	1.76±0.11
SplaTAM	0.35±0.05	0.34±0.06	0.27±0.05	0.19±0.09	0.32±0.07	0.31±0.01
Co-SLAM	1.75±0.28	1.58±0.03	1.47±0.03	1.51±0.05	1.55±0.02	2.05±0.49
ORB-SLAM3	71.54±0.66	66.49±0.86	65.16±2.77	66.59±1.76	69.99±0.79	69.06±0.67
DG-SLAM	1.22±0.15	1.14±0.00	1.41±0.01	1.32±0.00	0.87±0.00	1.11±0.01
RGD-SLAM	0.68±0.01	0.64±0.01	0.91±0.00	0.87±0.01	0.47±0.00	0.41±0.00
WildGS-SLAM	0.35±0.01	0.32±0.03	0.34±0.03	0.32±0.02	0.37±0.03	0.41±0.01
DynaGSLAM	0.60	0.61±0.00	0.68	0.62±0.00	0.70±0.00	0.72±0.00
DynaSLAM	6.29±3.12	5.18±2.37	3.02±1.02	2.70±0.82	13.63±8.98	7.72±4.20
NGD-SLAM	55.27±1.08	49.83±0.53	53.39±0.28	50.88±0.58	60.31±2.75	58.80±2.67
RoDyn-SLAM	1.63±0.02	1.63±0.03	1.68±0.03	1.68±0.04	1.65±0.03	1.62±0.03

Table S4: End-to-end frame rate (FPS, mean±sd over 3 seeds). Same-machine measurements under the reproduction protocol, not a throughput benchmark.

	near-lossless		*mid*		*aggressive*	
Sequence	ratio	PSNR	ratio	PSNR	ratio	PSNR
balloon	5.89×	15.33	8.64×	15.31	14.93×	14.89
balloon2	5.91×	14.89	8.56×	14.88	14.65×	14.59
crowd	6.02×	21.83	8.87×	21.58	15.29×	19.32
crowd2	6.12×	13.28	9.22×	13.32	16.14×	13.13
crowd3	6.05×	13.90	9.05×	13.88	15.69×	13.64
person_tracking	5.85×	13.47	8.54×	13.44	14.56×	13.32
person_tracking2	5.86×	14.90	8.51×	14.84	14.48×	14.30

Table S5: Codec working points per sequence (mean over the sequence's maps: 2 arms × 3 seeds). Bit widths are the fixed working points (16/8/8/8/8, 10/6/6/6/6, 8/4/4/4/4); only the per-map quantisation ranges are fitted per map. PSNR is the absolute full-image value; held-out-panel values are in the main text (§5).

Configuration	*Split*	n	ratio	B/prim	PSNR
SOG (JXL, unquantised)	L2 (frozen maps)	42	1.81×	44.1	−0.00
SOG (JXL, quantised)	L2 (frozen maps)	42	6.53×	12.3	−0.02
SOG (JXL, unquantised)	L3 (held-out)	36	1.81×	44.3	+0.00
SOG (JXL, quantised)	L3 (held-out)	36	6.61×	12.2	−0.04
Niedermayr c3dgs (QAT 5k)	L2 (frozen maps)	42	6.15×	13.1	16.15
Niedermayr c3dgs (QAT 5k)	L3 (held-out)	36	6.19×	13.1	15.89

Table S6: Learned-compressor detail. SOG's PSNR column is its change from the source (ΔPSNR); Niedermayr's is the absolute rendered PSNR (its refit seeds nothing, so per-map values are not reproducible beyond ≈0.1 dB and no Δ is claimed). Bookkeeping fields that SOG does not carry are compressed losslessly (lzma-9) and counted in the numerator; the denominator is unchanged. Working-point comparison against the parameter-free codec is in Table 4 of the main paper.

Sequence	*Arm*	bad PSNR gain	bad dL1 ratio	static Δ	full Δ	dyn. frac
balloon2	baseline	17.92	0.162	+0.33	+5.39	0.132
balloon2	final	18.79	0.153	+0.24	+6.31	0.114
crowd	baseline	22.22	0.185	+0.29	+12.01	0.239
crowd	final	22.22	0.187	+0.20	+11.91	0.222
crowd2	baseline	21.55	0.179	+0.04	+10.61	0.225
crowd2	final	21.54	0.176	−0.03	+11.04	0.286
crowd3	baseline	21.00	0.187	+0.39	+10.43	0.281
crowd3	final	21.31	0.182	+0.34	+10.26	0.218
person_tracking	baseline	19.51	0.269	+0.28	+8.70	0.062
person_tracking	final	18.12	0.238	+0.22	+8.41	0.101

Table S7: Per-map anti-dynamic cost (one row per frozen map). "bad PSNR gain" is the rendered-quality cost in the erased regions; "bad dL1 ratio" is the residual depth error there as a fraction of the erased value under a static insertion; "static Δ" and "full Δ" are rendered changes (dB) on the static and full image; "dyn. frac" is the erased-area pixel fraction. The main text reports the interval rather than a mean because the static side is sign-mixed.

Cell	J1 (dB)	J2	J3 (dB)	pass
balloon (seed 0)	8.95	0.230	$+0.10$	✓
balloon (seed 1)	7.89	0.270	$+0.20$	✓
balloon (seed 2)	8.64	0.233	$+0.12$	✓
balloon2 (seed 0)	6.08	0.227	$+0.07$	✓
balloon2 (seed 1)	6.40	0.282	$+0.04$	✓
balloon2 (seed 2)	5.76	0.339	$+0.03$	✓
crowd (seed 0)	5.76	0.581	$+0.03$	✓
crowd (seed 1)	5.85	0.579	$+0.00$	✓
crowd (seed 2)	5.49	0.610	$+0.04$	✓
crowd2 (seed 0)	7.95	0.473	-0.04	✓
crowd2 (seed 1)	7.47	0.520	-0.01	✓
crowd2 (seed 2)	7.73	0.450	-0.01	✓
crowd3 (seed 0)	7.89	0.444	$+0.03$	✓
crowd3 (seed 1)	7.52	0.464	$+0.04$	✓
crowd3 (seed 2)	7.56	0.505	$+0.03$	✓
person_tracking (seed 0)	7.38	0.427	$+0.00$	✓
person_tracking (seed 1)	7.24	0.406	$+0.00$	✓
person_tracking (seed 2)	7.67	0.412	$+0.00$	✓
person_tracking2 (seed 0)	6.42	0.550	$+0.00$	✓
person_tracking2 (seed 1)	6.82	0.478	$+0.00$	✓
person_tracking2 (seed 2)	7.18	0.461	$+0.00$	✓

Table S8: Offline repair, per sequence and seed: J1 = dynamic-region PSNR gain over the erased baseline; J2 = residual depth error as a fraction of the erased value; J3 = static-region change. Samepose medians; locked thresholds J1 ≥ 2.5 dB, J2 ≤ 0.70, J3 within ± 0.25 dB. Passed cells: 21/21.